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Esercizio 3.

a.

$$F = qvB \sin \theta = 8.4 \times 10^{-6} \text{ C} \cdot 45 \text{ m/s} \cdot 0.30 \text{ T} \cdot \sin(30^\circ) \approx 5.7 \times 10^{-5} \text{ N}$$

b.

$$F = qvB \sin \theta = 8.4 \times 10^{-6} \text{ C} \cdot 45 \text{ m/s} \cdot 0.30 \text{ T} \cdot \sin(90^\circ) \approx 1.1 \times 10^{-4} \text{ N}$$

c.

$$F = qvB \sin \theta = 8.4 \times 10^{-6} \text{ C} \cdot 45 \text{ m/s} \cdot 0.30 \text{ T} \cdot \sin(150^\circ) \approx 5.7 \times 10^{-5} \text{ N}$$

Esercizio 4.

$$B = \frac{F}{|q|v \sin \theta} = \frac{5.4 \times 10^{-3} \text{ N}}{|-8.3 \times 10^{-6} \text{ C}| \cdot 7.4 \times 10^6 \text{ m/s} \cdot \sin(52^\circ)} \approx 1.1 \times 10^{-5} \text{ T}$$

Esercizio 5.

$$B = \frac{F}{qv \sin \theta} \Rightarrow v = \frac{F}{qB \sin \theta} = \frac{5.9 \times 10^{-20} \text{ N}}{1.6 \times 10^{-19} \text{ C} \cdot 3.4 \times 10^{-5} \text{ T}} \approx 1.1 \times 10^4 \text{ m/s}$$

Esercizio 6.

$$B = \frac{F}{qv \sin \theta} \Rightarrow \sin \theta = \frac{F}{qvB} = \frac{4.6 \times 10^{-7} \text{ N}}{5.3 \times 10^{-9} \text{ C} \cdot 4.9 \times 10^4 \text{ m/s} \cdot 3.1 \times 10^{-3} \text{ T}} \approx 0.57$$

Applichiamo la funzione inversa e ricaviamo l'angolo acuto:

$$\theta = \arcsin(\sin \theta) = \arcsin(0.57) \approx 35^\circ$$